

LECTURE-02

Wednesday
19/8/26.□ Asymptotic Analysis:

Big-oh (O): Provides upper bound/worst case
 $f(n) = O(g(n))$ IF $f(n) \leq C * g(n)$

Big-Omega (Ω): Provides lower Bound/best case
 $f(n) = \Omega(g(n))$ IF $f(n) \geq C * g(n)$

Big-Theta (Θ): provides tight Bound (same func. provides upper & lower Bnd)
 $f(n) = \Theta(g(n))$ IF $C_1 * g(n) \leq f(n) \leq C_2 * g(n)$
 Average case

NOTE: $n \geq n_0$ where ' n_0 ' and ' c ' are +ve constants.

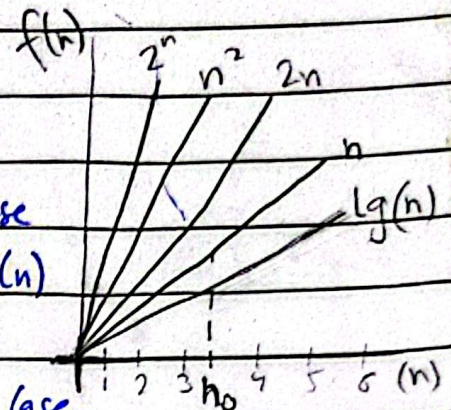
$$\textcircled{1}: 2n^2 + n = f(n)$$

n	f(n)	$\frac{2}{3} \cdot n^2 \cdot g(n)$	$\frac{1}{2} \cdot n^2 \cdot g(n)$
1	3	3	2
2	10	12	8
3	21	27	18
4	36	48	32
5	55	75	50

$f(n) = O(3n^2)$ ←
 as $f(n) \leq g(n) * c$
 where $c = 3$ and
 $n_0 \geq 1$

$f(n) = \Omega(2n^2)$
 as $f(n) \geq g(n) * c$
 where $c = 2$ and
 $n_0 \geq 1$

$\therefore f(n) = \Theta(n^2)$
 with $C_1 = 2$ and
 $C_2 = 3$ with
 $n_0 \geq 1$



$$Q: 4n+4 = f(n)$$

$$f(n) = O(n) \text{ with } c_1 = 4 \text{ and } c_2 = 8 \text{ with } n_0 \geq 1.$$

* In Big-Theta; n_0 is a single value $\therefore$ we can't take $C_2 = 5$ (even tho it is a tighter upper bound) as $C_2 = 5$ will make our func(RHS) " $5n$ " which is not an upper bound for $n = 1$ (which should be true as we gave " $n_0 \geq 1$ " for lower bound).